Elastic deformation

From stress–strain curves to rock properties

How much does a rock deform, and how much of that deformation can it recover?

Learning objectives and lecture sequence

By the end of this lecture, you should be able to:

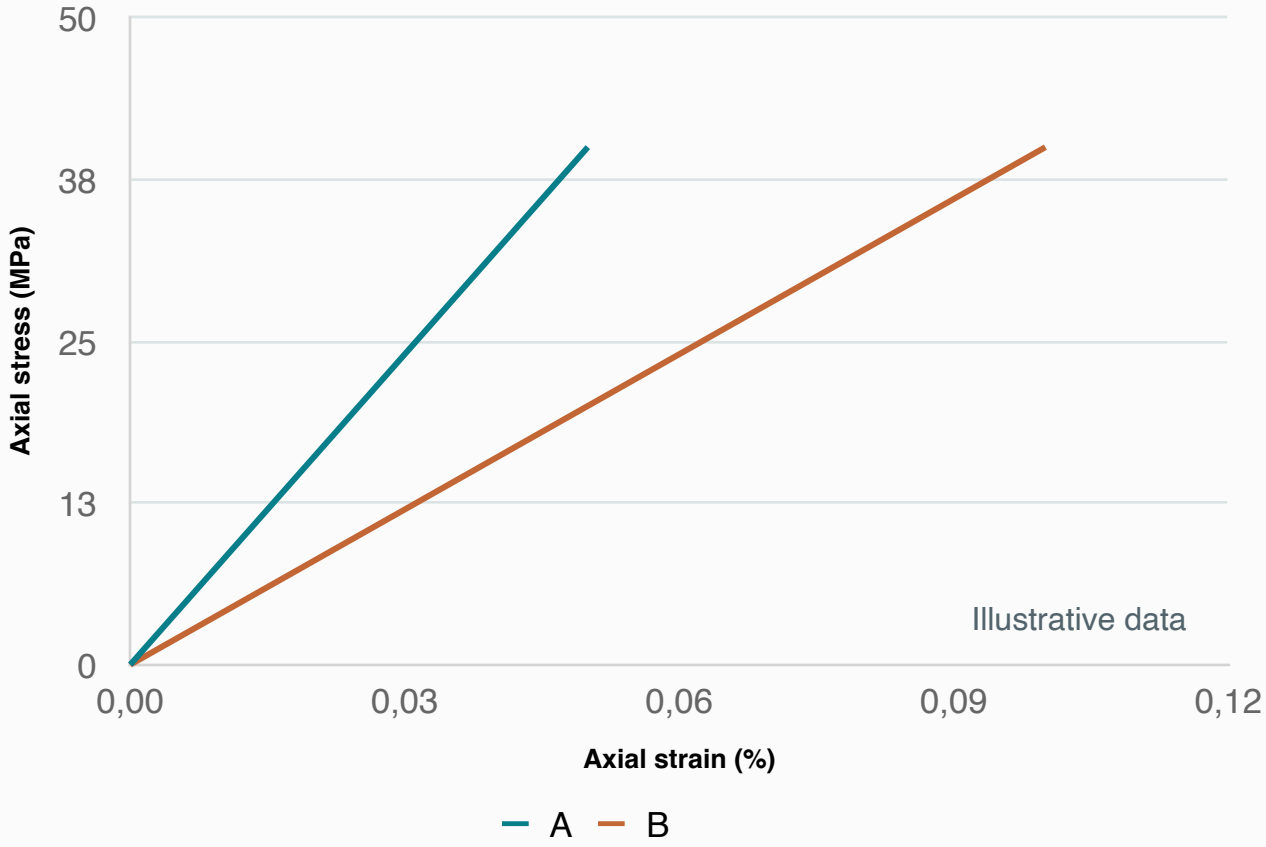
Learning objective	Main evidence or calculation
Distinguish elastic and permanent deformation	Loading and unloading paths
Determine Young’s modulus and Poisson’s ratio	Axial and lateral strain measurements
Predict deformation under simple loading	Uniaxial stress, pressure and shear
Evaluate an elastic approximation for a rock	Choice of interval and measurement method

Opening question: two specimens under the same stress

Two specimens have the same dimensions. At an axial stress of 40 MPa, specimen A shortens by 0.05% and specimen B by 0.10%.

Which specimen is stiffer? Explain your reasoning using the stress–strain plot.

Can these measurements also tell us which specimen will fail at a higher stress?



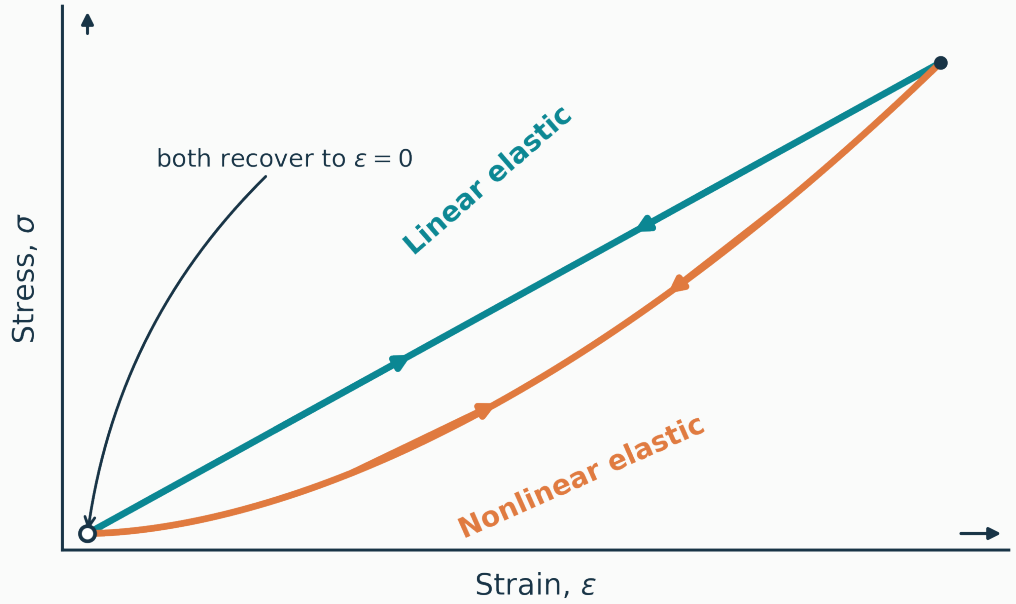
Think individually, then compare explanations with a neighbour.

Elasticity describes recoverable deformation

Elastic deformation disappears when the applied stress returns to its initial value. An ideal elastic material stores mechanical energy during loading and returns it during unloading.

Linear elasticity adds a second condition: stress is proportional to strain. A material can recover its deformation even when this relationship is nonlinear.

Permanent strain remains after unloading. A loading curve alone cannot show how much strain will recover.



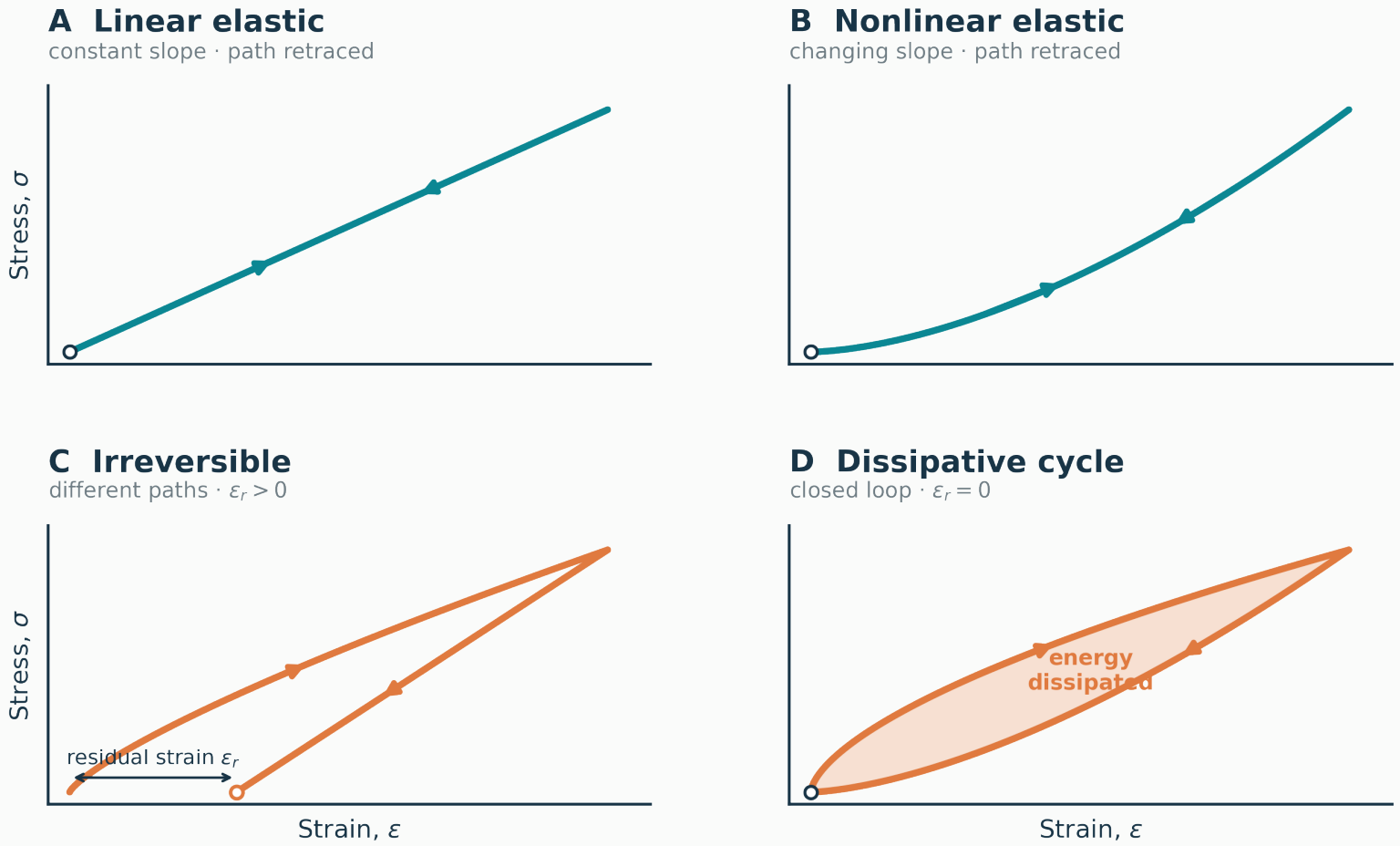
Recovery and linearity are different properties of the response.

Loading and unloading reveal different behaviours

A **linear elastic** path has a constant slope. A **nonlinear elastic** path can still retrace the loading curve completely.

The illustrated **irreversible** response unloads to a positive residual strain. Its loading and unloading paths differ.

A **loop** indicates mechanical energy dissipation. A loop can occur even when the final residual strain is small or zero.



Stiffness and strength answer different questions

Property	Definition	What determines it in a test?
Stiffness/Moduli	Resistance to deformation	Stress–strain slope in a stated interval
Strength	Stress sustained at a specified failure condition	Peak or another defined failure criterion
Elastic recovery	Return toward the initial dimensions	Strain measured during unloading

A high elastic modulus does not, by itself, specify a high failure stress. We need separate measurements of the deformation response and the failure condition.

The opening example ranks stiffness, but leaves strength undetermined.

Sign convention and model assumptions

We follow Paterson and Wong: compressive stress and shortening strain are positive. An increase in length gives a negative normal strain.

$$\varepsilon_a = (L_0 - L) / L_0 \qquad \varepsilon_r = (D_0 - D) / D_0$$

For the next equations, assume small strains, a homogeneous response and isotropic linear elasticity. Isotropy means that the elastic response has no preferred direction.

Uniaxial stress means that only the axial stress is nonzero. The lateral surfaces are free to expand. We neglect end effects in this idealisation.

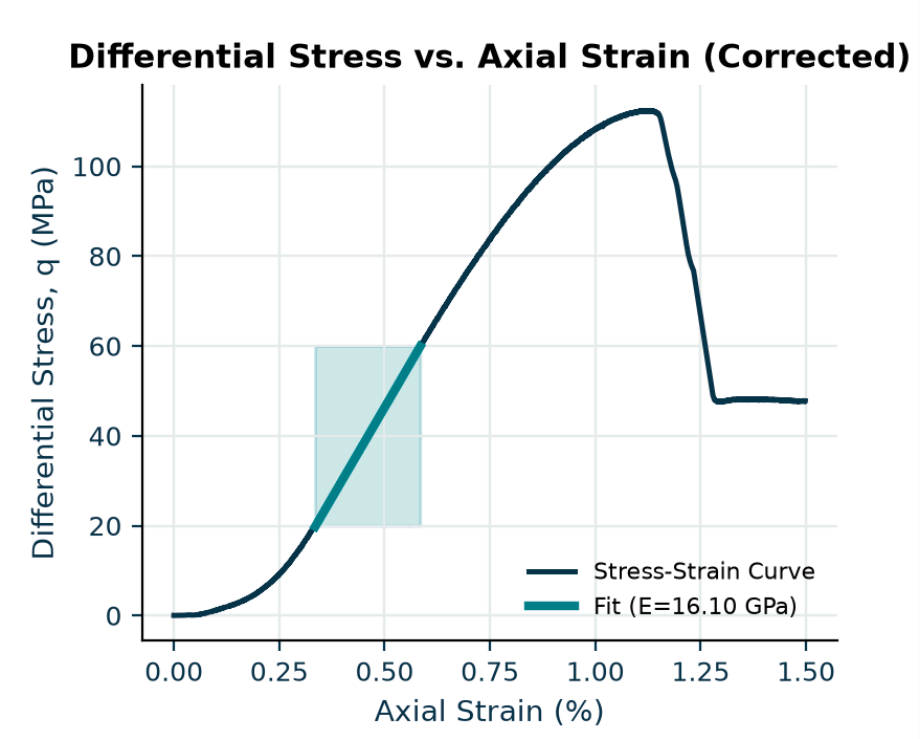
Young's modulus under uniaxial stress

Activity 1

Young's modulus E relates axial stress to axial strain when the lateral stresses are zero. A larger E gives a smaller axial strain at the same stress.

$$\sigma_{axial} = E * \epsilon_a \qquad E = \frac{\sigma_{axial}}{\epsilon_a}$$

Strain is dimensionless, so E has units of stress. We commonly report stress in MPa and Young's modulus in GPa.





$\sigma_1 > 0$

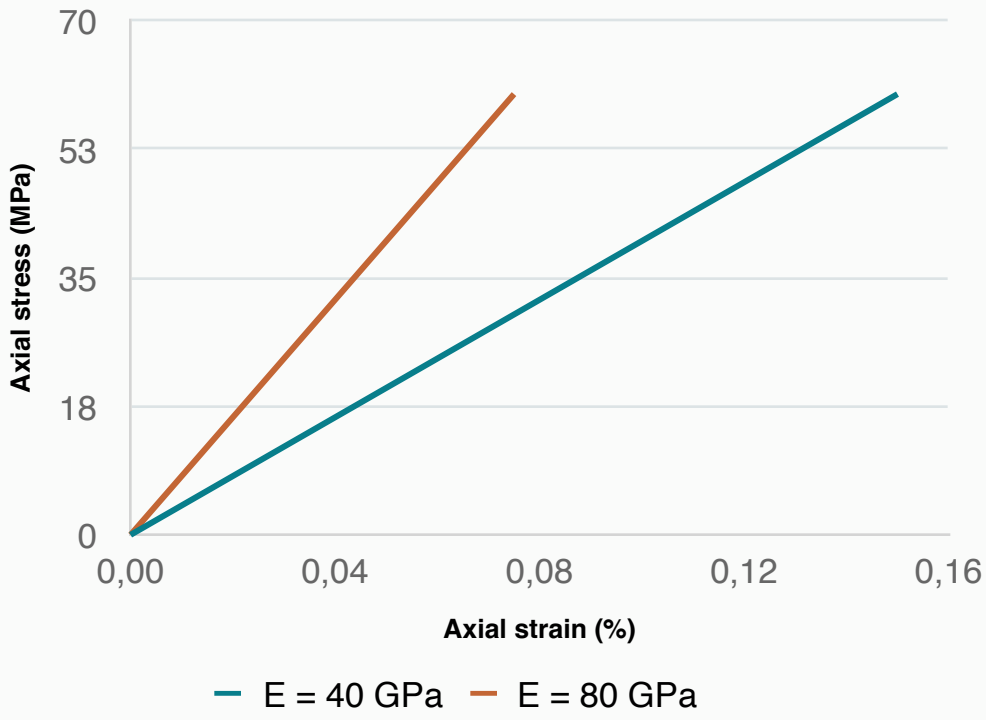
$\sigma_2 = \sigma_3 = 0$

Young’s modulus is the stress–strain slope

For an ideal linear elastic material, the stress–strain curve is a straight line. Doubling E doubles its slope.

Here both specimens experience the same stress. At 40 MPa, the material with E = 80 GPa develops half the strain of the material with E = 40 GPa.

$$\varepsilon_a = \sigma_a / E$$



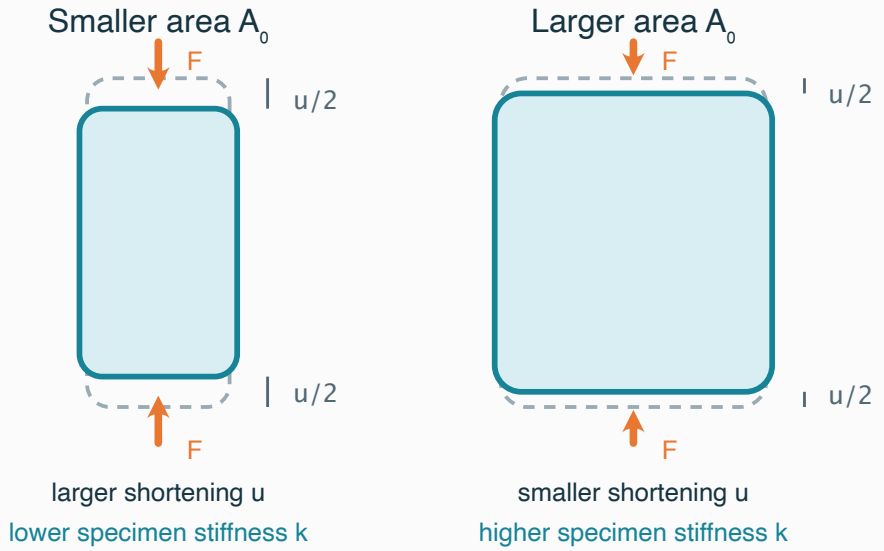
Illustrative data

When strain is plotted as %, convert it to a fraction before calculating E.

Specimen stiffness and material modulus

The force needed to shorten a specimen depends on its **material and its geometry**.
Let u be positive shortening, A_0 the initial area and L_0 the initial length.

$$F = \frac{EA_0}{L} * u \qquad k = \frac{F}{u} = \frac{EA_0}{L_0}$$



A wider specimen is harder to shorten, even when the material modulus is unchanged.

Stress and strain remove the first-order effect of specimen geometry.



Paterson & Wong (2005) §2.1.4, pp. 13–14

Worked example: shortening of a cylindrical sample

Consider an ideal uniaxial test with $E = 40$ GPa, initial length $L_0 = 100$ mm and diameter $D_0 = 40$ mm. Apply an axial stress of 40 MPa.

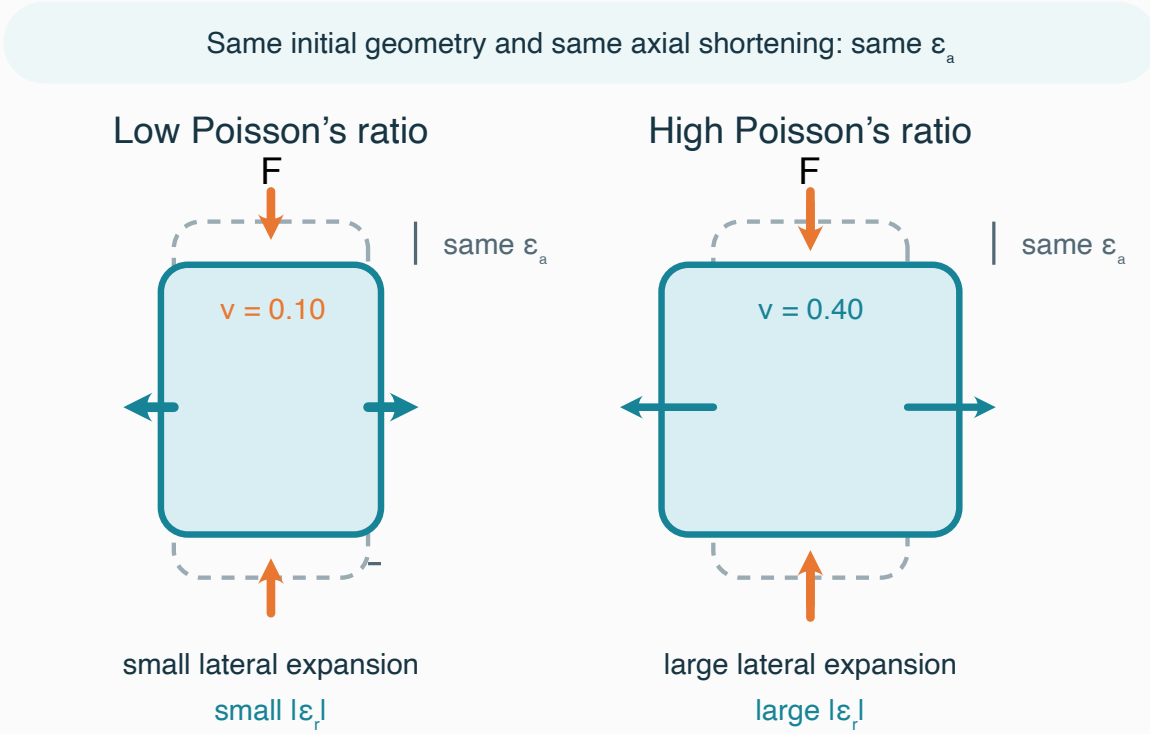
Step	Calculation	Result
Axial strain	$\varepsilon_a = 40 / 40\,000$	0.001 = 0.10%
Shortening	$u = \varepsilon_a L_0$	0.10 mm
Initial area	$A_0 = \pi D_0^2 / 4$	1256.6 mm²
Axial force	$F = \sigma_a A_0$	50.3 kN

Poisson’s ratio describes lateral coupling

Axial compression usually produces lateral expansion in an ordinary elastic rock. Poisson’s ratio v measures the magnitude of this coupling under uniaxial stress.

$$\nu = -\frac{\epsilon_r}{\epsilon_a}$$

With compression positive, axial shortening gives $\varepsilon_a > 0$ and lateral expansion gives $\varepsilon_l < 0$. The minus sign therefore gives a positive v for this response.



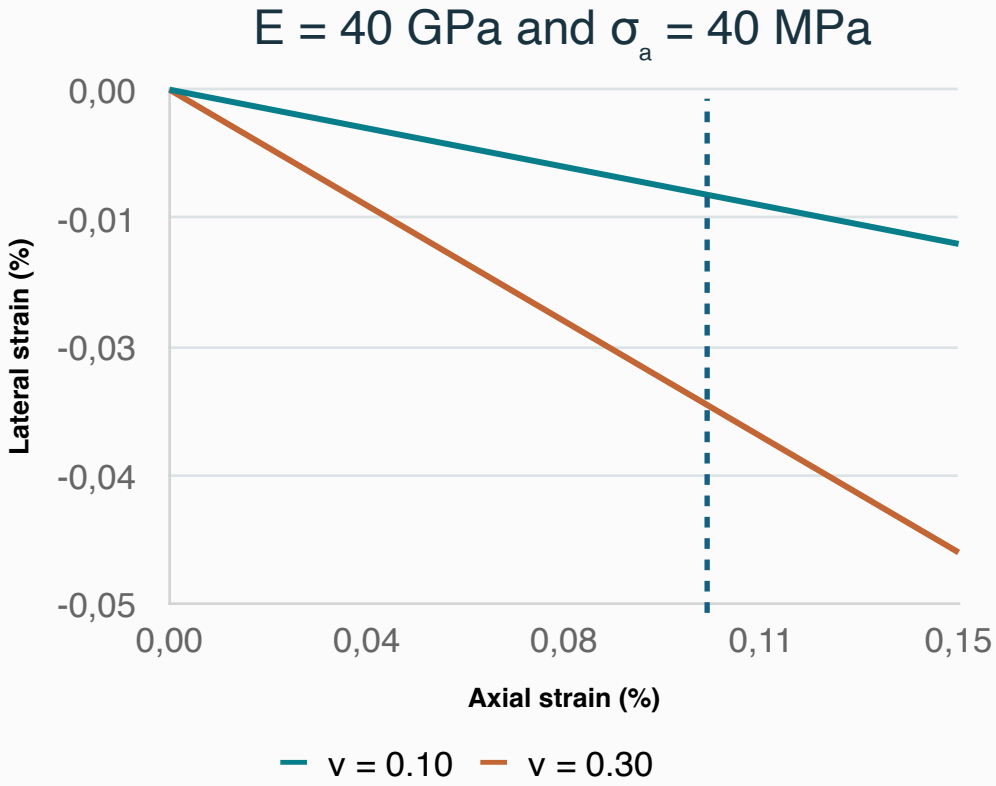
E controls axial strain. At fixed axial strain, v controls lateral strain.

Lateral expansion at different Poisson’s ratios

Keep E = 40 GPa and σ_a = 40 MPa. The axial strain remains 0.10% while the lateral response changes with v.

For v = 0.10, lateral strain is –0.01%. For v = 0.30, it is –0.03%.
Both specimens shorten equally, but one widens more.

Question: does greater lateral expansion necessarily imply an increase in total volume?



Illustrative data

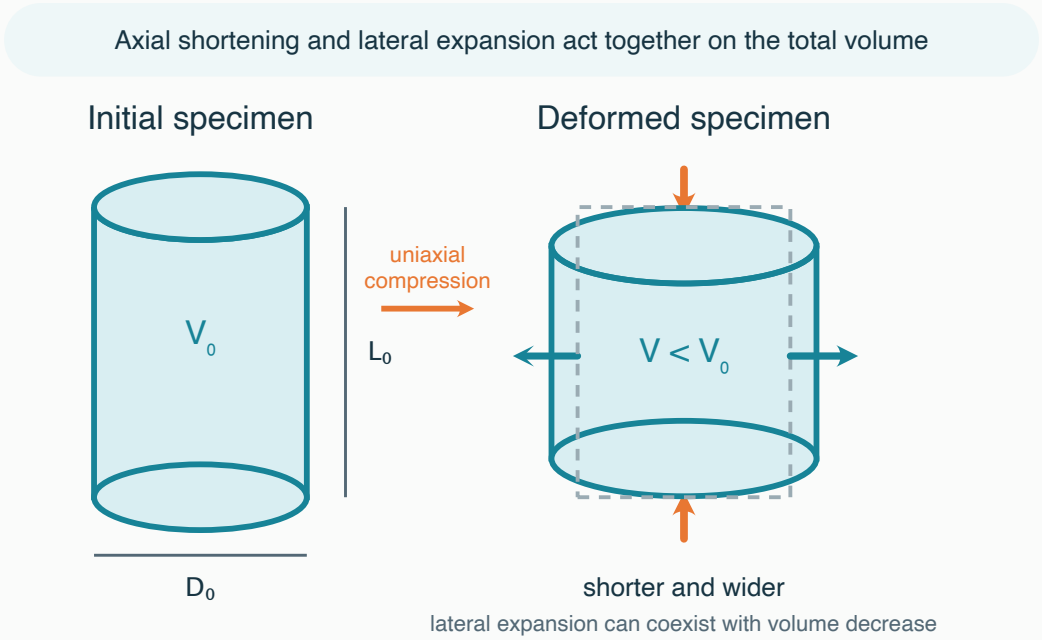
Volumetric strain combines all normal strains

At small strain, the fractional volume change follows the sum of the three normal strains. Positive volumetric strain means a decrease in volume.

$$\epsilon_v = \frac{V_0 - V}{V_0} = \epsilon_1 + \epsilon_2 + \epsilon_3$$

For **axisymmetric deformation**, the two lateral strains are equal. Under isotropic linear elasticity and uniaxial stress:

$$\epsilon_v = \epsilon_a + 2\epsilon_r = (1 - 2\nu)\epsilon_a$$

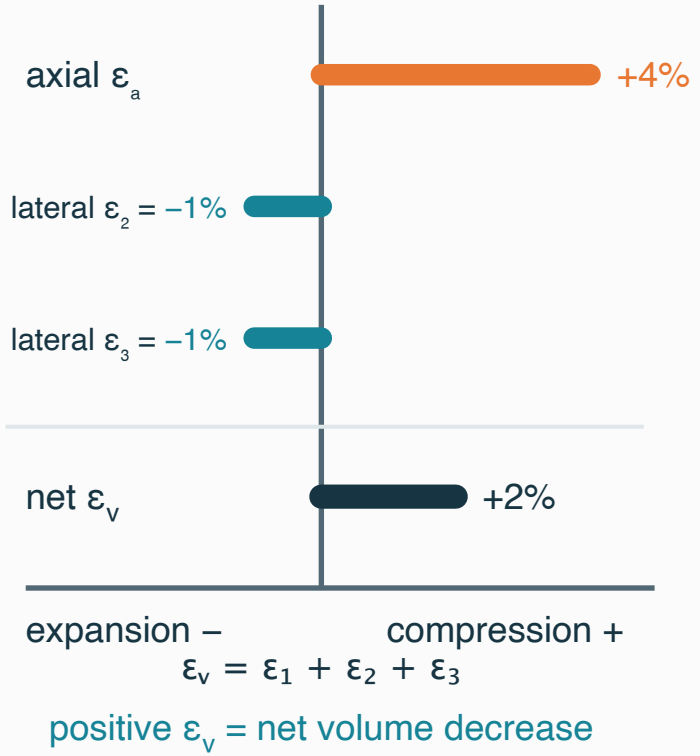


Lateral expansion and an overall volume decrease can occur together.

Volumetric strain combines all normal strains

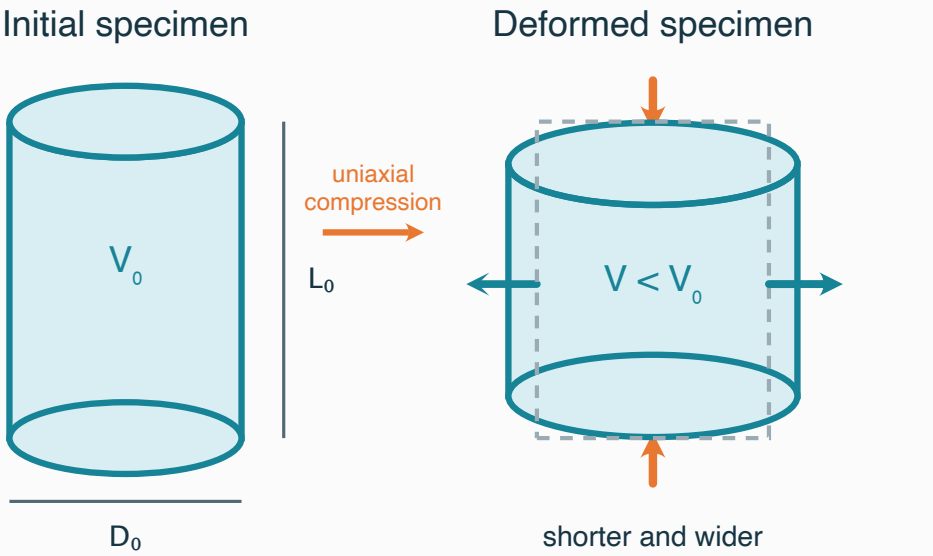
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$$\epsilon_v = \epsilon_a + 2\epsilon_r = (1 - 2\nu)\epsilon_a$$



positive ε_v = net volume decrease

Axial shortening and lateral expansion act together on the total volume



Lateral expansion and an overall volume decrease can occur together.

Paterson & Wong (2005) §5.3, pp. 68–70

Worked example: widening while volume decreases

Return to the 100 mm long, 40 mm diameter sample. With E = 40 GPa, v = 0.25 and σ_a = 40 MPa:

Quantity	Calculation	Physical meaning
Axial strain	$\varepsilon_a = 0.001$	Length decreases by 0.10 mm
Lateral strain	$\varepsilon_l = -0.25 \times 0.001$	Diameter increases by 0.010 mm
Volumetric strain	$\varepsilon_v = 0.001 - 0.00050$	Volume decreases by about 0.05%

The axial contribution exceeds the two lateral contributions combined. A wider specimen can therefore occupy less total volume.

Bulk modulus: resistance to volume change

Activity 2

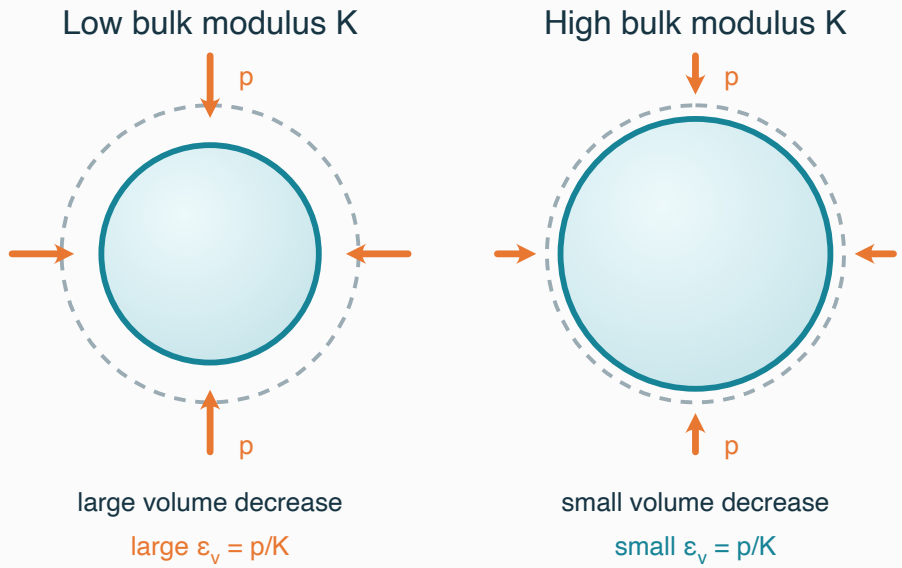
Under hydrostatic compression, all principal stresses equal the pressure p. The specimen contracts equally in every direction if its response is isotropic.

$$\sigma_1 = \sigma_2 = \sigma_3 = p \qquad K = \frac{p}{\epsilon_v}$$

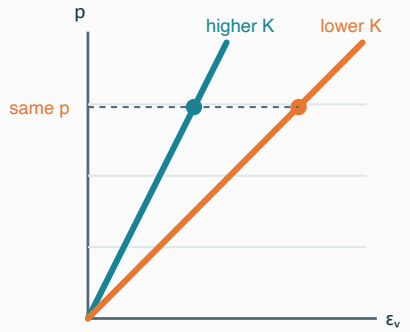
The bulk modulus K measures resistance to elastic volume change. Compressibility is its reciprocal, 1/K. A larger K gives a smaller volume decrease at the same pressure.

$$K = \frac{E}{3(1-2\nu)}$$

Same initial volume V_0 and same hydrostatic pressure p



K is the slope p/ε_v



steeper line = smaller volume change

The shape is preserved; only its scale changes.

Shear modulus: resistance to shear deformation

Activity 3

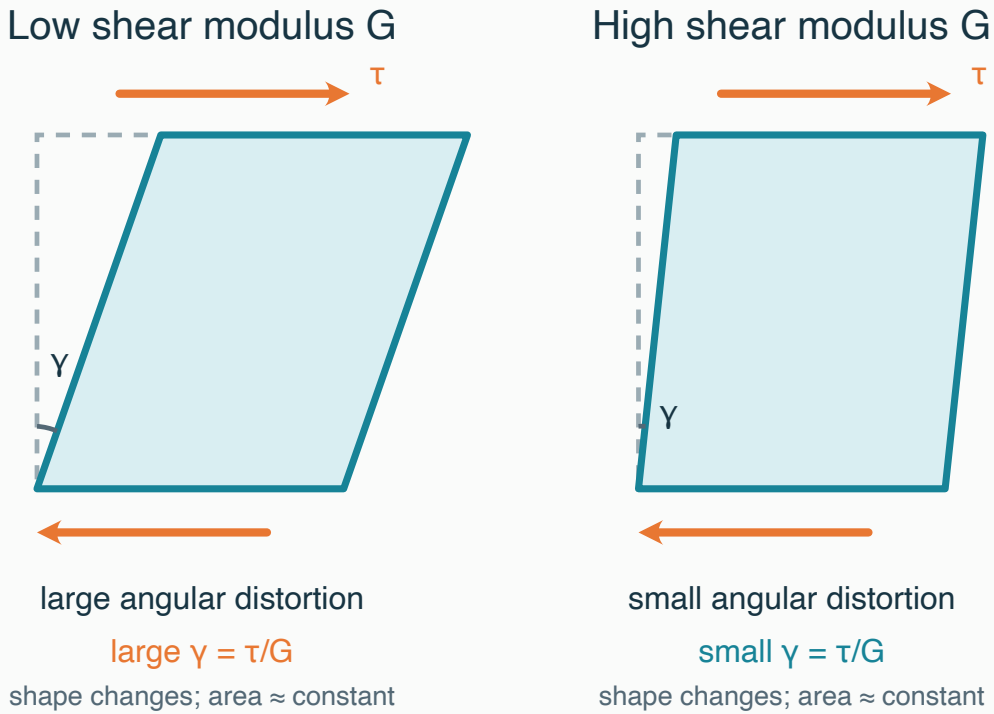
Pure shear changes shape without a first-order volume change. The shear modulus G relates shear stress τ to engineering shear strain γ.

$$\tau = G\gamma \qquad \gamma = 2\varepsilon_{12}$$

Engineering shear strain is the small change of angle between initially perpendicular material lines, in radians. Tensor shear strain ε_{12} is half this quantity.

$$G = \frac{E}{2(1+\nu)}$$

Same initial shape and same applied shear stress τ



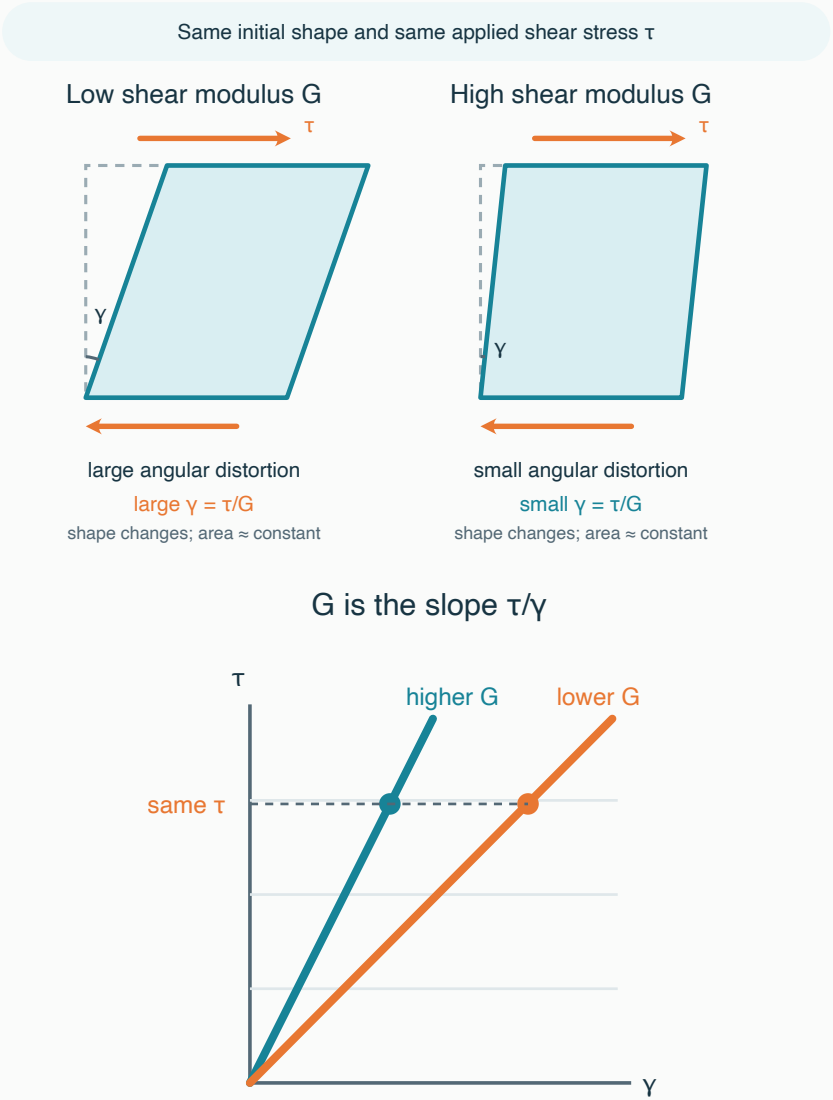
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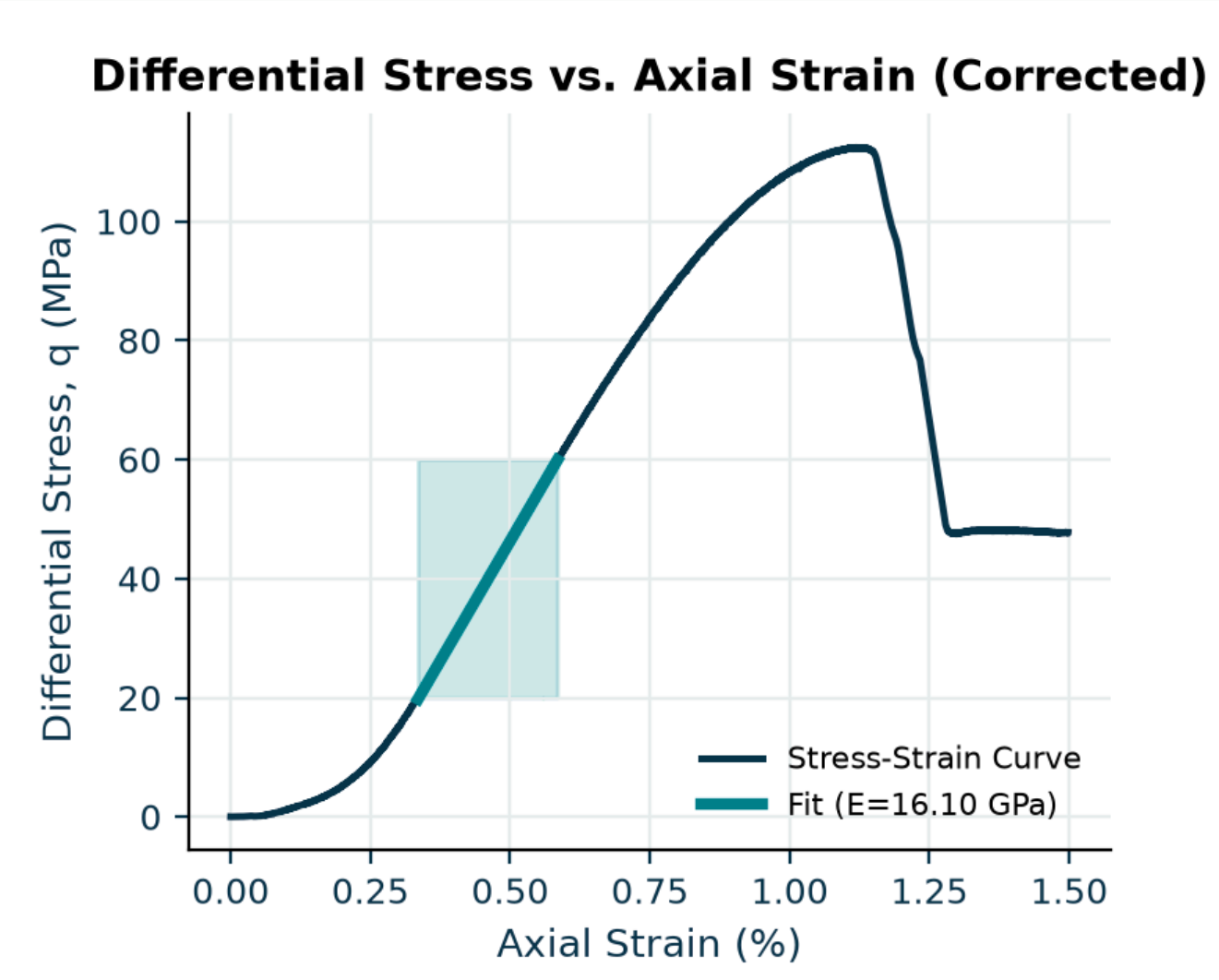


Two independent constants describe isotropic elasticity

Quantity	Loading or deformation	Relation
Young’s modulus E	Uniaxial stress and axial strain	$E = \sigma_a / \varepsilon_a$
Poisson’s ratio v	Lateral coupling in uniaxial stress	$v = -\varepsilon_l / \varepsilon_a$
Bulk modulus K	Hydrostatic pressure and volume strain	$K = E / [3(1 - 2v)]$
Shear modulus G	Shear stress and engineering shear strain	$G = E / [2(1 + v)]$

For E = 40 GPa and v = 0.25: K = 26.7 GPa and G = 16.0 GPa. These relations assume isotropic linear elasticity. An anisotropic rock needs additional elastic constants.

Why real rocks depart from the ideal model



Paterson & Wong (2005) §5.3, pp. 70–71; §5.4.1, pp. 76–77

What we can now infer from deformation

- Stress and strain describe the loading and response. Elastic moduli connect them through a material law with explicit assumptions.
- Axial and lateral strain together tell us about stiffness, coupling and elastic volume change. Loading and unloading help distinguish recovery from permanent deformation.
- In a real rock, the stress interval, crack state and measurement method influence the result. These details become essential when we interpret the compression experiment.

Feedback welcome



Next: how elastic waves probe the same rock through a different measurement.

Reading

Paterson, M. S. & Wong, T.-f. (2005). Experimental Rock Deformation: The Brittle Field, 2nd edition, Springer.

Reading in the book	Connection to this lecture
Preface, p. VI	Compression-positive stress and strain
§2.1.4, pp. 13–14	Force, displacement and apparatus corrections
§5.3, pp. 68–71	Volumetric strain and an elastic reference
§5.4.1, pp. 76–77	Elastic properties and measurement approaches